

RRGE-RAG: Radiology Report Generation Enhancement Using Hybrid Retrieval Augmented Generation

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Abstract

Automated radiology report generation is an important application of medical artificial intelligence, aiming to reduce radiologists' workload while improving diagnostic efficiency. In this progress report, we present the implementation and ongoing evaluation of our proposed Hybrid Retrieval-Augmented Generation (Hybrid RAG) framework for chest X-ray report generation. The proposed framework integrates visual features, structured clinical knowledge, and retrieved case-based evidence to enhance the factual correctness and coherence of generated reports. We implement and fine-tune several baseline models using their original source code to ensure fair comparison on the Indiana University Chest X-ray dataset. Our approach extends the R2Gen architecture by incorporating BioViL-T as the image encoder and introducing a feature alignment and attention-based fusion mechanism to combine visual embeddings with retrieved reports and clinical labels. For evaluation, we employ lexical metrics (BLEU, ROUGE-L), clinical metrics (CheXbert F1, RadGraph F1), and uncertainty-aware metrics, including Expected Calibration Error (ECE) and Brier Score. Furthermore, we refine our uncertainty-focused research question to analyze the relationship between uncertainty estimates and generation quality, as well as model calibration. At the current stage, the implementation of baseline models, the Hybrid RAG framework, and the evaluation pipeline has been completed. Experimental evaluations and comparative analyses are ongoing. This report outlines the system design, implementation progress, and planned evaluation toward developing a reliable and clinically applicable radiology report generation system.

automatically produce a textual report describing the clinical findings visible in the images. (Jing et al., 2018) However, medical report writing requires specialized domain expertise, and only experienced radiologists can accurately interpret chest X-ray images and describe the corresponding findings in a coherent manner. ¹An automatic chest X-ray report generation system can reduce the workload of radiologists and are in urgent need (Brady et al., 2012). Recent multimodal foundation models have exhibited remarkable capabilities in challenging healthcare tasks, motivating an automation of this process to enhance physicians' efficiency on clinical decision-making and improve patient health outcomes. (Sun et al., 2025)

Although prior medical multimodal foundation models have demonstrated promising capabilities on report generation given the radiology image, they still suffer from serious hallucinations by generating factually inaccurate reports (Pal et al., 2023; Pal and Sankarasubbu, 2024; Ahmad et al., 2023). Factual correctness is especially critical in chest radiology domains, as minute textual differences can drastically invert radiology report meaning and downstream prescribed treatments (Delbrouck et al., 2022; Song et al., 2024; Liu et al., 2024). Retrieval-Augmented Generation (RAG) has emerged as a popular paradigm to address this issue by grounding text generation with retrieved relevant knowledge given a query (Lewis et al., 2020; Wolf et al., 2024; Gao et al., 2023). However, developing medical multimodal retrievers remains challenging, requiring retrievers to bridge the gap between symptomatic image semantics and factually-equivalent report text.

1 Introduction

A major medical application of natural language generation (NLG) is the development of assistive systems that analyze patient X-ray images and

To capture fine-grained details in chest ra-

¹<https://github.com/Vibolrottanaseng/NLU-Project-Wave2Vec>

diographs and improve the factual completeness of generated reports we introduce **RRGE-RAG**. It is a framework that takes as input a patient’s chest X-ray images, including both frontal and lateral views, together with structured clinical labels derived from the Problems column. Given these inputs, the system performs retrieval to identify the Top-3 most similar prior patient cases based on image similarity. Each retrieved case provides complementary contextual information, including the corresponding frontal and lateral chest X-ray images, the associated radiology report composed of Findings and Impression sections, and the structured clinical labels. By leveraging these retrieved examples as external evidence, the model is guided to generate a clinically grounded report for the input case. The primary generation target is the Impression section, since it summarizes the key diagnostic conclusion in a concise and clinically useful form, while generation of the Findings section can be considered as an optional extension. This design allows the framework to combine visual evidence from the current patient with relevant knowledge from similar historical cases, with the goal of improving factual accuracy, reducing hallucination, and producing more reliable radiology reports as shown in Figure 1.

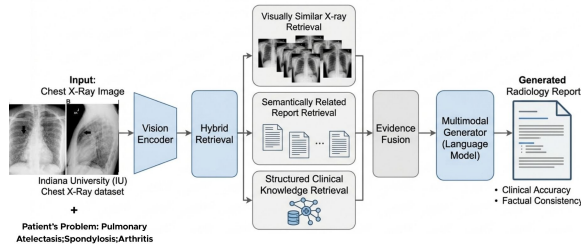


Figure 1: An Overview of RRGE-RAG framework. Hybrid RAG framework for chest X-ray report generation. A chest X-ray from the Indiana University dataset is passed through a vision encoder, then a hybrid retrieval module retrieves three types of supporting evidence: visually similar X-rays, semantically related reports, and structured clinical knowledge. These retrieved evidences are combined through evidence fusion and sent to a multimodal language generator, which produces the final radiology report with improved clinical accuracy and factual consistency.

We have came up with 3 research question for this study:

1. How does the integration of retrieval-

augmented generation (RAG) influence the accuracy and clinical relevance of automated radiology report generation compared to traditional image-to-text models? This question evaluates whether adding retrieval improves report generation beyond standard image-to-text models such as R2Gen. Traditional models rely only on learned image-text mappings, which can lead to incomplete or hallucinated findings. By retrieving similar prior cases, the proposed method aims to ground generation in relevant external evidence. This question can be answered by comparing the proposed model with baseline methods using BLEU and ROUGE-L, while also examining whether retrieval leads to more clinically relevant and factually consistent reports.

2. To what extent does multimodal fusion of X-ray images, clinical labels, and retrieved case information improve the coherence and completeness of generated radiology reports? This question focuses on the effect of combining multiple information sources in the proposed framework. The model integrates image features, structured clinical labels, and retrieved case information to provide richer context for report generation. This can be examined through ablation studies that compare image-only input with versions that add labels, retrieval, and full multimodal fusion. Better performance from the full model would indicate that combining these modalities improves report coherence and completeness.

3. How well do uncertainty estimates (measured using entropy and MC dropout variance) correlate with generation quality (BLEU, ROUGE), and how well are they calibrated (ECE, Brier Score)? This question examines whether the model’s uncertainty estimates are meaningful and reliable. If higher entropy or MC dropout variance is associated with lower BLEU or ROUGE scores, then uncertainty can serve as an indicator of poor-quality generation. Calibration metrics such as ECE and Brier Score further assess whether the model’s confidence matches its actual performance. This is important for clinical settings, where confidence information can help determine whether a generated report should be trusted or reviewed more carefully.

We expect that integrating retrieval mechanisms and multimodal fusion will significantly improve the quality of generated reports. Specifically, the proposed approach is anticipated to produce more accurate, coherent, and clinically relevant reports by leveraging contextual knowledge from similar cases. Additionally, the uncertainty estimation module is expected to provide meaningful confidence scores that correlate with the correctness of the generated outputs. Overall, the system should outperform traditional approaches in both quantitative metrics (e.g., BLEU, ROUGE, ECE, Brier Score) and qualitative clinical assessment.

2 Experimental Settings

This section describes the overall methodology of the proposed system for automated radiology report generation. The pipeline follows a multimodal, retrieval-augmented framework that integrates image features, clinical knowledge, and retrieved case information. An overview of the system is illustrated in Figure 1, which presents the end-to-end flow from input data to report generation and evaluation.

2.1 Dataset

We use the **Indiana University Chest X-ray dataset**² for this study. This dataset consists of chest X-ray images and corresponding radiology reports for **3,851 patients**. For each patient, two views are typically available: a *frontal view* and a *lateral view*, which provide complementary anatomical information for clinical interpretation. In addition to images, the dataset includes a CSV file (`indiana_reports.csv`) containing structured and unstructured clinical information. The main columns used in this work are:

- **Problems:** A set of clinically relevant findings identified from the X-ray, representing medical conditions or abnormalities (e.g., cardiomegaly, effusion). These are derived from expert-written radiology reports.
- **Image:** Metadata describing the type or view of the X-ray (e.g., PA, lateral).
- **Indication:** The clinical reason for performing the X-ray (e.g., symptoms or suspected condition).

²<https://www.kaggle.com/datasets/raddar/chest-xrays-indiana-university>

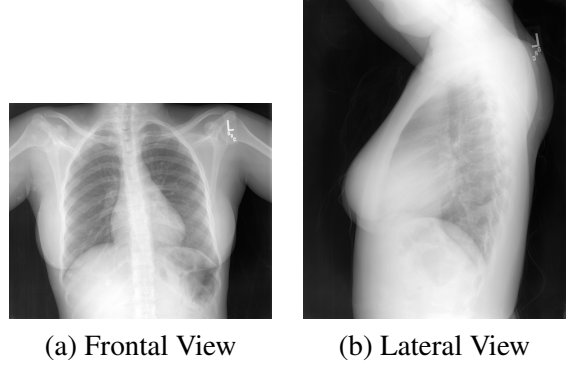


Figure 2: Example chest X-ray images from the IU dataset showing (a) frontal and (b) lateral views for a single patient.

- **Comparison:** Information about prior studies used for comparison, if available.
- **Findings:** A detailed descriptive section of the radiology report explaining observations in the X-ray.
- **Impression:** A concise summary of the most important clinical conclusions derived from the findings.

For our methodology, the inputs consist of **two X-ray images (frontal and lateral views)** along with the corresponding **Problems** (clinical labels), which provide structured medical knowledge. The target output is the **Findings** section, which serves as the ground truth radiology report to be generated by the model.

2.2 Data Preprocessing

Data preprocessing is a critical step to ensure consistency, data quality, and effective integration of multimodal inputs within the proposed **Hybrid Retrieval-Augmented Generation (Hybrid RAG)** framework. To maintain fairness across all experiments, the same preprocessing pipeline is applied to both baseline models and the proposed method.

Image Preprocessing For each patient, chest X-ray images are organized to include both *frontal* and *lateral* views, which provide complementary anatomical information. All images are resized and normalized according to the input requirements of the BioViL-T vision encoder. These standardized images are then used to extract domain-specific visual embeddings for both feature learning and retrieval.

Text Preprocessing For report generation, the *Findings* section is used as the target output. The text is preprocessed by converting to lowercase, removing unnecessary special characters, and normalizing whitespace while preserving clinically relevant terms. A word-level tokenizer is applied to construct the vocabulary, and special tokens such as <start>, <end>, and <pad> are introduced. Each report is then converted into a sequence of token IDs and padded or truncated to a fixed maximum length. For text-based retrieval, reports are minimally preprocessed and encoded into dense semantic representations using a biomedical language model (e.g., BiomedBERT). This enables retrieval based on semantic similarity rather than exact lexical matching.

Structured Clinical Label Processing The *Problems* column is used to extract structured clinical knowledge. Each entry may contain multiple conditions, which are separated and converted into a multi-hot encoded vector representing the presence or absence of each condition. These structured labels are incorporated both as model inputs and as an additional signal in the hybrid retrieval process.

Data Filtering To ensure data quality and consistency, samples that do not meet the following criteria are excluded:

- (i) missing or empty *Problems* labels,
- (ii) absence of either frontal or lateral images, and
- (iii) missing or invalid *Findings* reports.

After filtering, the final dataset consists of 3,337 samples with aligned image, text, and structured label information.

Dataset Splitting and Consistency The dataset is divided into training (70%), validation (10%), and test (20%) sets at the patient level to prevent data leakage. Same train, validation, and test splits are used across all baseline models and the proposed Hybrid RAG framework. This ensures a fair and consistent evaluation by eliminating variability due to different data distributions. As a result, any observed performance differences can be attributed to the model architectures and methodological contributions rather than inconsistencies in data partitioning.

2.3 Baselines

To evaluate the effectiveness of the proposed **Hybrid Retrieval-Augmented Generation (Hybrid**

RAG) framework, we compare it with several established baseline models:

- **R2Gen** (Chen et al., 2020): A standard image-to-text model for radiology report generation.
- **R2GenCMN** (Chen et al., 2021): An extension of R2Gen that incorporates cross-modal memory for improved generation.
- **PPKED** (Liu et al., 2021): A knowledge-enhanced model that integrates external medical knowledge.
- **METransformer** (Wang et al., 2023): A transformer-based architecture with multiple expert tokens.

These baselines represent diverse categories of approaches, including standard, memory-based, knowledge-enhanced, and modern transformer-based architectures, enabling a comprehensive and fair comparison with the proposed method.

Our proposed approach builds upon **R2Gen as the report generation backbone** and extends it with a **hybrid Retrieval-Augmented Generation (RAG) framework**. For image representation and retrieval, we employ **BioViL-T** (Bannur et al., 2023) as the image encoder to extract domain-specific visual embeddings from chest X-rays. For textual retrieval, we utilize **BiomedBERT (PubMedBERT-style) sentence embeddings** (Gu et al., 2021) to capture semantic similarity between radiology reports.

2.4 Evaluation Metrics

To ensure a comprehensive evaluation of the proposed framework, we employ three categories of metrics: lexical, clinical, and uncertainty-aware calibration metrics. These metrics jointly assess linguistic quality, clinical correctness, and prediction reliability.

Lexical Metrics We use standard text generation metrics to evaluate the similarity between generated and reference reports:

- **BLEU (BLEU-2, BLEU-3, BLEU-4)** (Papineni et al., 2002): Measures n-gram overlap between generated and ground truth reports at different granularities.
- **ROUGE-L** (Lin, 2004): Captures structural similarity by computing the longest common

subsequence between generated and reference texts.

Clinical Metrics While lexical metrics evaluate surface-level similarity, they do not guarantee clinical correctness. Therefore, we employ clinically grounded evaluation metrics:

CheXbert F1 Score. (Smit et al., 2020) CheXbert (?) is used to convert generated reports into structured disease labels. We consider 14 clinically relevant chest conditions, commonly used in prior work (?):

- Atelectasis, Cardiomegaly, Consolidation, Edema, Effusion, Emphysema, Fibrosis, Hernia, Infiltration, Mass, Nodule, Pleural Thickening, Pneumonia, Pneumothorax

From these labels, we compute four variants of F1 scores:

- **14Ma-F1 (14-class Macro F1):** Computes F1 independently for each of the 14 diseases and averages them equally. This treats all diseases equally, including rare conditions.
- **14Mi-F1 (14-class Micro F1):** Aggregates predictions across all 14 labels and computes a global F1 score. This is influenced more by frequent diseases.

The use of 14-label settings enables evaluation across both *comprehensive disease coverage* and *high-frequency clinical conditions*, ensuring balanced assessment of model performance.

RadGraph F1 Score (RG-F1). RadGraph (Jain et al., 2021) evaluates the correctness of extracted medical entities (e.g., observations and anatomical locations) and their relationships. Unlike CheXbert, which focuses on classification, RadGraph captures the structural and relational correctness of clinical information in generated reports.

Together, CheXbert and RadGraph provide a robust evaluation of **clinical accuracy**, ensuring that generated reports are not only fluent but also medically meaningful.

Uncertainty and Calibration Metrics In clinical settings, it is essential not only to produce accurate predictions but also to estimate their reliability. To evaluate model uncertainty, we employ the following metrics:

Expected Calibration Error (ECE). ECE (Posocco and Bonnefoy, 2021) measures how well predicted confidence scores align

with actual correctness. Predictions are grouped into M confidence bins, and the difference between average confidence and accuracy is computed:

$$ECE = \sum_{m=1}^M \frac{|B_m|}{N} |acc(B_m) - conf(B_m)|$$

where B_m is the set of samples in bin m , $acc(B_m)$ is the accuracy, and $conf(B_m)$ is the average confidence.

A lower ECE indicates better calibration, meaning the model’s confidence reflects true correctness.

Brier Score. (Redelmeier et al., 1991) The Brier Score measures the accuracy of probabilistic predictions by computing the mean squared error between predicted probabilities and ground truth labels:

$$Brier = \frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2$$

In our setting, each generated report is converted into 14 binary disease labels using CheXbert. The model produces a probability p_i for each label, and the ground truth $y_i \in \{0, 1\}$. The Brier Score is computed across all labels and averaged over the dataset.

This formulation adapts naturally to the multi-label classification setting, allowing us to evaluate how well the predicted probabilities reflect true clinical outcomes.

Interpretation for Uncertainty Evaluation ECE and Brier Score provide complementary insights into model uncertainty:

- Low ECE indicates that confidence estimates are well calibrated.
- Low Brier Score indicates accurate probability predictions.
- High ECE with high accuracy suggests *miscalibration* (e.g., underconfidence or overconfidence).

These metrics enable us to assess whether the model’s uncertainty estimates are meaningful and reliable, which is critical for clinical decision-support applications.

2.5 Method Overview

Figure 1 illustrates the overall architecture of the proposed system. The framework follows a retrieval-augmented multimodal pipeline that integrates visual features, textual knowledge, and structured clinical information to generate accurate radiology reports.

The system performs **multi-signal hybrid retrieval** by combining image-based similarity, text-based similarity, and structured clinical labels (Problems). Retrieved items are treated as **case-level units** (image, report, and labels), ensuring clinical consistency. These multimodal features are integrated through an **attention-based fusion mechanism**, which dynamically learns the importance of different modalities and retrieved cases. The fused representation is then passed to the R2Gen generator to produce the final radiology report.

2.6 Implementation Detail

The proposed **Hybrid Retrieval-Augmented Generation (Hybrid RAG)** framework extends a standard image-to-text pipeline by integrating multimodal retrieval and attention-based fusion to improve report quality and clinical reliability.

Visual Encoding and Feature Alignment The BioViL-T encoder extracts visual embeddings from input chest X-ray images, producing feature representations:

$$V \in R^{N \times d}$$

Since these embeddings differ from the feature space expected by R2Gen, a linear projection layer is applied to map them into the latent space of the R2Gen model:

$$V' = WV$$

where W is a learnable transformation matrix ensuring dimensional compatibility.

Hybrid Retrieval To incorporate external knowledge, the model retrieves top-3 similar cases based on:

- visual similarity (image embeddings),
- semantic similarity (report embeddings),
- structured clinical labels (multi-hot vectors).

Each retrieved case consists of image features, report text, and clinical labels, preserving cross-modal consistency.

Multimodal Fusion and Generation The projected visual features (V'), retrieved representations (R), and label embeddings (L) are combined using an attention-based fusion mechanism:

$$F = \text{Attention}(V', R, L)$$

The fused representation F serves as the encoder output for the R2Gen decoder, which generates the radiology report in a sequence-to-sequence manner.

Training Configuration The model is trained using cross-entropy loss and optimized with the Adam optimizer. To ensure stable and effective training, we apply teacher forcing, dropout, gradient clipping, and early stopping based on validation performance.

This implementation enables effective integration of multimodal inputs and retrieved knowledge, leading to more accurate, coherent, and clinically grounded report generation.

2.7 Training Strategy

The proposed model is trained in a supervised manner using paired chest X-ray images and corresponding radiology reports. The **Findings** section serves as the ground truth target for report generation.

Dataset Split The preprocessed dataset of 3,337 samples is divided into three subsets:

- **Training set (70%)**: Used to learn model parameters.
- **Validation set (10%)**: Used for hyperparameter tuning and model selection.
- **Test set (20%)**: Used for final performance evaluation.

The split is performed at the patient level to avoid data leakage between sets.

Training Procedure Each sample includes frontal and lateral X-ray images, structured labels, and the target *Findings* report. Images are encoded using **BioViL-T**, and top-3 similar cases are retrieved based on image, text, and label similarity. Retrieved and input features are fused via attention and passed to the **R2Gen** decoder for sequence generation using teacher forcing.

Loss and Optimization The model is trained using **cross-entropy loss** and optimized with the **Adam optimizer**. Early stopping based on validation performance is applied.

Validation and Regularization Model performance is monitored using validation loss and BLEU scores. **Dropout**, **gradient clipping**, and **teacher forcing** are used to improve stability and generalization.

3 Results and Analysis

3.1 Comparison with Baselines

As shown in Table 1, different models exhibit strengths across lexical and clinical dimensions. In terms of lexical performance, **METransformer** achieves the best results on BLEU-1 (**0.483**), BLEU-3 (**0.228**), and BLEU-4 (**0.172**), demonstrating strong fluency and n-gram consistency. Meanwhile, **R2GenCMN** obtains the highest ROUGE-L score (**0.545**), indicating superior structural alignment with reference reports. **R2Gen** performs competitively on BLEU-2 (**0.479**).

For clinical evaluation, **METransformer** achieves the highest RG-F1 score (**0.311**), highlighting its effectiveness in capturing medical entities and their relationships. In contrast, **R2GenCMN** achieves the best CheXbert-based scores, with **0.097** in 14Ma-F1 and **0.717** in 14Mi-F1, indicating strong performance in disease classification, particularly for frequent conditions.

The proposed **Hybrid RAG** model achieves moderate lexical scores (BLEU-4: 0.084, ROUGE-L: 0.299) while maintaining competitive RG-F1 performance (0.2535), comparable to R2Gen and PPKED. This suggests that retrieval-based augmentation helps preserve clinically relevant information, although it may reduce surface-level similarity.

Overall, these results highlight a trade-off between lexical fluency and clinical grounding, where transformer-based models excel in text generation, while retrieval-based approaches aim to enhance factual consistency.

3.2 Ablation Study

The ablation results in Table 2 highlight the contribution of each component. The baseline model (A0: R2Gen) achieves the best performance in BLEU-1 (**0.479**), BLEU-2 (**0.479**), ROUGE-L (**0.483**), and 14Mi-F1 (**0.667**), indicating strong lexical and classification performance.

In contrast, the A1 configuration, which relies only on structured clinical labels, shows a significant degradation in lexical metrics (e.g., BLEU-4: 0.001), confirming that label information alone is

insufficient for coherent report generation. However, it achieves the highest 14Ma-F1 (**0.078**), suggesting some benefit in rare disease classification.

The proposed Hybrid RAG model improves substantially over A1, achieving the best BLEU-3 (**0.113**), BLEU-4 (**0.084**), and RG-F1 (**0.253**) within the ablation setting. This demonstrates that combining retrieval with multimodal fusion is essential for balancing linguistic quality and clinical relevance.

3.3 Qualitative Analysis

From a qualitative perspective, transformer-based models such as METransformer generate fluent and structurally consistent reports, which contributes to their strong lexical scores. However, these models may rely heavily on learned patterns, potentially limiting their ability to incorporate case-specific clinical evidence.

In contrast, the Hybrid RAG model leverages retrieved cases, enabling it to incorporate contextually relevant clinical information. This improves clinical grounding, as reflected in RG-F1 performance, but may introduce variability in phrasing, leading to lower BLEU and ROUGE scores.

This observation suggests a fundamental trade-off between **lexical similarity** and **clinical correctness**, where retrieval-based approaches prioritize factual consistency over exact textual matching.

4 Discussion

The integration of retrieval-augmented generation (RAG) improves clinical grounding and factual consistency, as reflected by stronger RG-F1 performance, although it leads to slightly lower lexical scores (BLEU and ROUGE) compared to traditional image-to-text models.

Multimodal fusion of X-ray images, structured clinical labels, and retrieved case information enhances the completeness and clinical relevance of generated reports, demonstrating clear improvements over using individual modalities alone.

Uncertainty estimates are expected to correlate with generation quality; however, further analysis using calibration metrics such as ECE and Brier Score is required to determine whether the model’s confidence is reliable and well-calibrated.

5 Conclusion

In this work, we proposed a **Hybrid Retrieval-Augmented Generation (Hybrid RAG)** frame-

Model	BL-1	BL-2	BL-3	BL-4	ROUGE-L	RG-F1	14Ma-F1	14Mi-F1
R2Gen	0.479	0.479	0.103	0.082	0.483	0.242	0.058	0.667
R2GenCMN	0.362	0.267	0.203	0.153	0.545	0.254	0.097	0.717
PPKED	0.099	0.052	0.032	0.021	0.178	0.247	0.040	0.423
METransformer	0.483	0.322	0.228	0.172	0.480	0.311	0.074	0.080
Proposed (Hybrid RAG)	0.288	0.165	0.113	0.084	0.299	0.2535	0.041	0.041

Table 1: Comparison of baseline models and the proposed Hybrid RAG framework. Best results are highlighted in bold.

Model	BL-1	BL-2	BL-3	BL-4	ROUGE-L	RG-F1	14Ma-F1	14Mi-F1
A0 (R2Gen)	0.479	0.479	0.103	0.082	0.483	0.242	0.058	0.667
A1 (+Labels)	0.058	0.014	0.004	0.001	0.055	0.011	0.078	0.376
Proposed Model	0.288	0.165	0.113	0.084	0.299	0.253	0.041	0.041

Table 2: Ablation study evaluating the contribution of different components. Best results are highlighted in bold.

work for radiology report generation, integrating visual features, structured clinical labels, and retrieved case-based knowledge.

Experimental results demonstrate that while strong transformer-based baselines achieve superior performance in lexical metrics, the proposed approach provides competitive clinical grounding, particularly in terms of entity-level correctness (RG-F1). The ablation study further confirms that retrieval and multimodal fusion are essential components, as neither structured labels nor baseline models alone achieve balanced performance.

Overall, this work demonstrates the potential of combining retrieval-based grounding with multimodal learning for radiology report generation. Future work will focus on improving retrieval quality, refining fusion strategies, and evaluating uncertainty of generated output to better align model confidence with prediction reliability, thereby enabling safer and more clinically applicable deployment.

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